

20/01

Single PHASE AC CKTS (MODULE-2)

⑧ ADVANTAGES OF AC OVER DC;

- 1.) ^{High} AC voltages are easier to generate upto 11 KV or more whereas in DC high voltages cannot be generated bcoz of presence of commutators (mechanical rectifiers).
- 2.) AC voltages can easily be stepped up using transformers, transmitted to long dist. & again step down which is not possible in DC.
- 3.) AC induction motors are simpler, cheaper & more rugged as compared to DC motors.

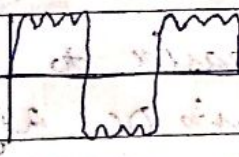
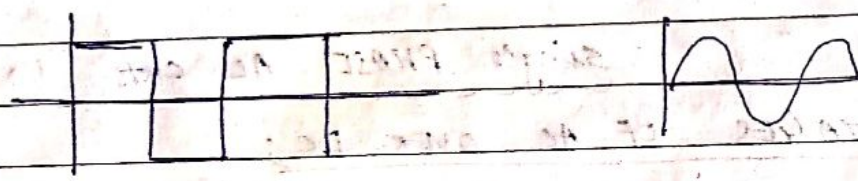
(like remotely operating something)
operation from low to high voltage

4.) The switch gears i.e. relays & circuit breakers in AC are simpler & cheaper as those in DC.

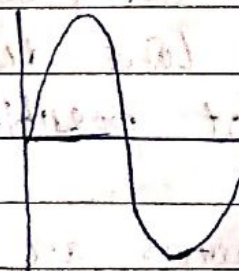
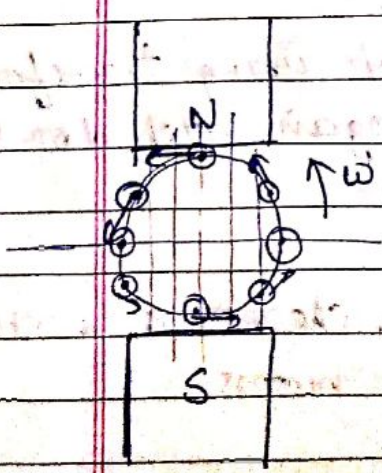
(*) ADVANTAGES OF SINE WAVE OVER OTHER AC FREQ.

- 1.) AC voltages are easier to generate
- 2.) Sum of 2 sinusoids having same frequency is also a sinusoid.
- 3.) Differentiation & Integration of Sinusoid is also a sinusoid.
- 4.) Any period waveform can be expressed as sum of sinusoids having diff. frequencies.
(Fourier series)

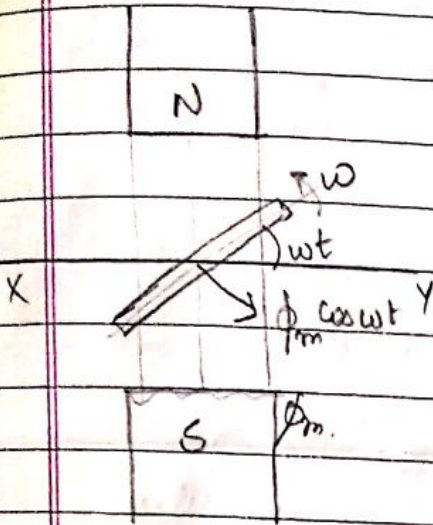
$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega t + b_n \sin n\omega t)$$



(*) GENERATION OF ALTERNATING VOLTAGE



⇒ Equation of Alternating Voltage :-



At any time 't' it makes an angle ωt with XY axis.
The component of the flux $\perp$ to the axis of the coil is given by $\phi_m \cos \omega t$.

Acc. to Faraday's law of EMI, emf induced.

$$e = -N \frac{d\phi(t)}{dt}$$

$$= N \phi_m \omega \sin \omega t$$

$$e = E_m \sin \omega t$$

⇒ Average or Mean Value of Alternating Current :-

It is that value of DC current which when passing through the same circuit for same time transfers same amt. of charge. $I_{avg} = \frac{\text{area of the curve}}{\text{length of base of curve}}$

⇒ Effective or Root Mean Square value of AC :-

is that value of DC current which when passing through the same circuit for same time produce ~~more~~ same amt. of heat.

⇒ Sine Wave.

$$I_{avg} = \frac{2 I_m}{\pi} \quad (\text{half cycle})$$

$$I_{rms} = \frac{I_m}{\sqrt{2}}$$

Also do derivations (from I rms)

Half rectified:-

$$I_{avg} = \frac{I_m}{\pi}$$

$$I_{rms} = \frac{I_m}{2}$$

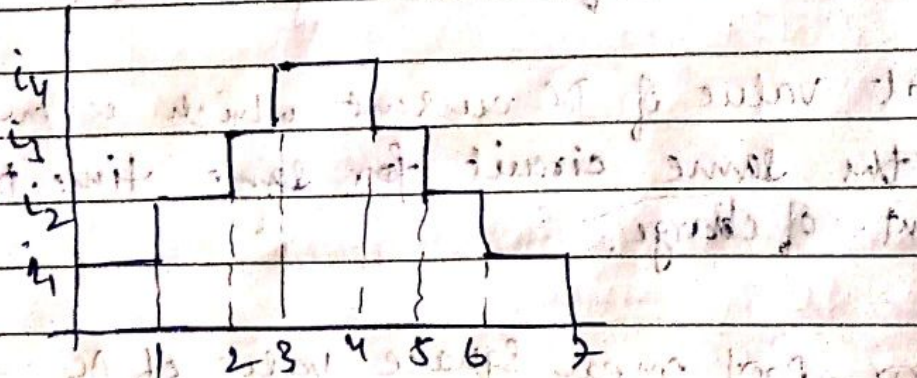
$$k_f \text{ (form factor)} = \frac{I_{rms}}{I_{avg}}$$

$$k_p \text{ (peak factor)} = \frac{I_m}{I_{rms}}$$

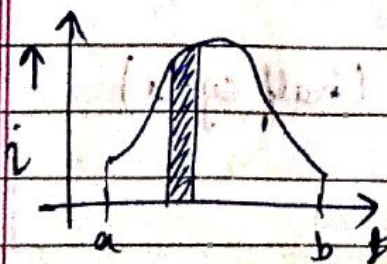
(*) Mid ordinate method:-

$$I_{avg} = \frac{i_1 + i_2 + i_3 + i_4 + i_5 + i_6 + i_7}{7}$$

$$I_{rms} = \sqrt{\frac{i_1^2 + i_2^2 + i_3^2 + i_4^2 + i_5^2 + i_6^2 + i_7^2}{7}}$$



(*) By Integration:- (Sine)



$$I_{avg} = \frac{\int_a^b i(t) dt}{(b-a)}$$

$$I_{rms} = \sqrt{\frac{\int_a^b i^2(t) dt}{(b-a)}}$$

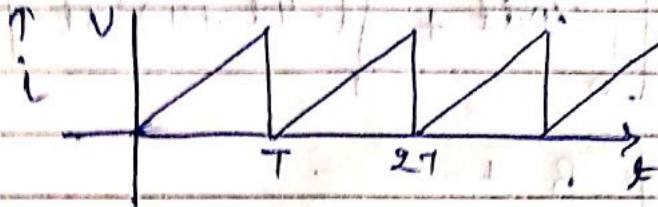
RMS value always more than avg. value.

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Q.

Find avg. & rms value of i :-

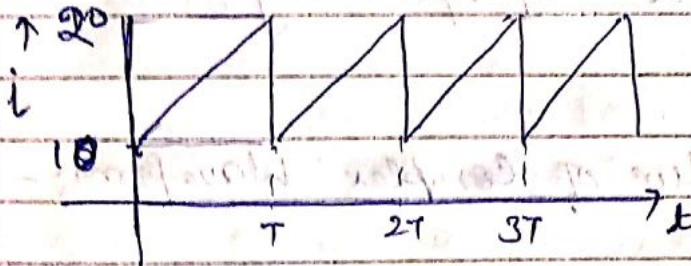


$$I_{avg} = \frac{\frac{1}{2} VT}{T} = \frac{V}{2}$$

$$i(t) = \frac{V}{T} t$$

$$I_{rms} = \sqrt{\frac{1}{T} \int_0^T \left(\frac{Vt}{T}\right)^2 dt} = \frac{V}{\sqrt{3}}$$

Q.



$$I_{avg} = \frac{10T + \frac{1}{2}10T}{T} = 15$$

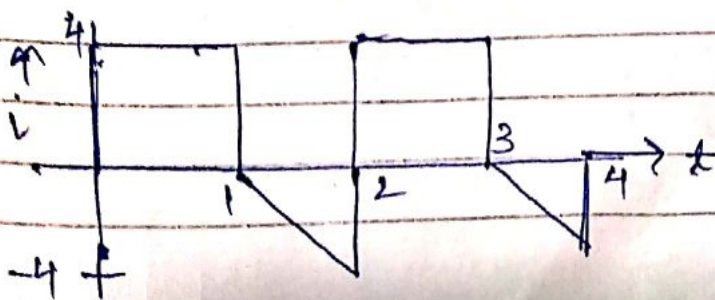
or $I_{avg} = \frac{1}{T} \int_0^T \left(10 + \frac{10t}{T}\right) dt$

$$I_{rms} = \sqrt{\frac{1}{T} \int_0^T \left(10 + \frac{10t}{T}\right)^2 dt}$$

$$= 15.27$$

$y = mx + c$
 $\frac{10}{T} t + 10$

Q.



$i = 4 ; 0 \leq t \leq 1$
 $i = -4t + 4 ; 1 \leq t \leq 2$

$$I_{avg} = \frac{4 \times 1 - \frac{1}{2} 4 \times 1}{2} = 1$$

$$I_{rms} = \sqrt{\frac{1}{2} \left[\int_0^1 (4)^2 dt + \int_1^2 (1-4t+4)^2 dt \right]}$$

$$= \sqrt{\frac{1}{2} [8 +$$

$$= 3.26$$

I_{rms} is much greater than avg. becoz -ve area is subtracted in avg. while in rms it becomes +ve by squaring.

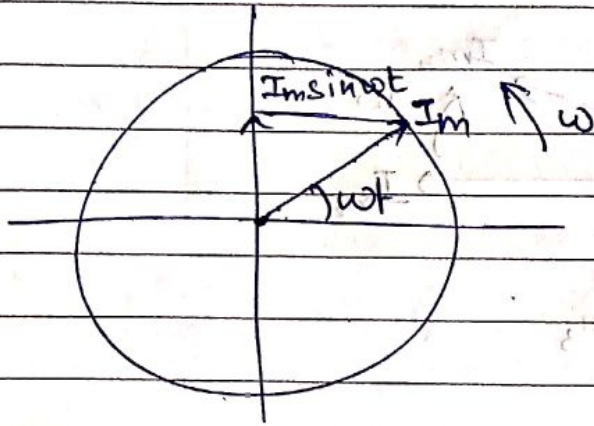
(*) RMS + AVG value of Complex waveform:-

Q. $i(t) = 200 + 45 \sin 3\omega t + 10 \sin 5\omega t$
Find avg. + rms value.

Solⁿ = $I_{avg} = 200$ becoz in 1 complete cycle value of sin is 0.

$$I_{rms} = \sqrt{200^2 + \left(\frac{45}{\sqrt{2}}\right)^2 + \left(\frac{10}{\sqrt{2}}\right)^2}$$

Phasors:- $\rightarrow$ rotating vector
 $\rightarrow$ only ~~repr~~ sinusoidal waveforms are represented:



$$e = E_m \sin \omega t.$$

Phasor is a st. line fixed at one end with its mag. equal to peak value of sinusoidal quantity rotating in anti-clockwise dir. with constant ang. speed ω .

At any instant of time its vertical projection gives the instantaneous value of the alternating quantity.

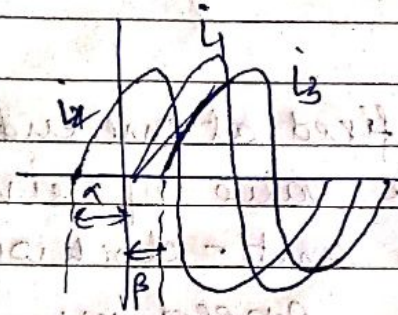
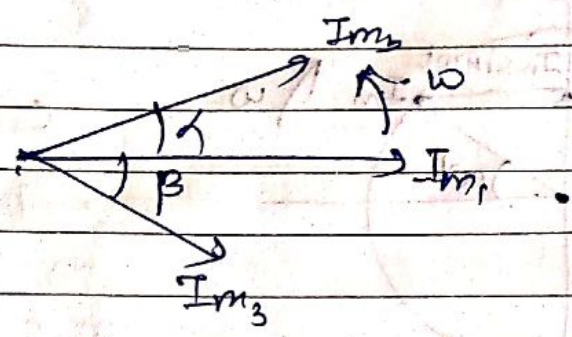
In practice more than one sinusoidal quantity may be involved in calculations. All these quantities may be represented on same diag. by drawing phasors inclined at diff. angles to refer. line.

If these phasors have same freq. they will rotate at same ω their phasors will change with time bt. phase diff. b/w any 2 phases will remain same.

$$i_1 = I_{m1} \sin \omega t$$

$$i_2 = I_{m2} \sin (\omega t + \alpha)$$

$$i_3 = I_{m3} \sin (\omega t - \beta)$$



Graphical representation of the phasors of sinusoidal quantities of same freq. showing proper phase relationship w.r.t each other is c/a phasor diagram.

Voltage & current phasors can be drawn on same phasor. bt. voltage can be added/sub. to voltage & current add to a current.

Polar Representation

$$\vec{I}_1 = |I_1| \angle 0^\circ$$

$$\vec{I}_2 = |I_2| \angle \alpha$$

$$\vec{I}_3 = |I_3| \angle -\beta$$

$$1000 (i) = j^9$$

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Rectangular Representation:-

$$\begin{aligned} I_1 \cos 0^\circ + j I_1 \sin 0^\circ \\ I_2 \cos \alpha + j I_2 \sin \alpha \\ I_3 \cos(-\beta) + j I_3 \sin(-\beta) \\ I_3 \cos \beta - j I_3 \sin \beta \end{aligned}$$

eg-

$$\begin{aligned} A \angle \alpha \times B \angle \beta \\ = AB \angle (\alpha + \beta) \end{aligned}$$

Polar.

$$\frac{A \angle \alpha}{B \angle \beta} = \frac{A}{B} \angle \alpha - \beta$$

$$i_1 = 3 + j4$$

$$i_2 = 4 + j7$$

⑧ Conversion of Rectangular to Polar

$$3 + j4$$

$$\text{Pol}(3, 4) = 5$$

then press alpha & press

F it gives angle.

$$= 5 \angle 53.13^\circ$$

Polar to Rectang.

Press shift + Rec (5, 53.13°)

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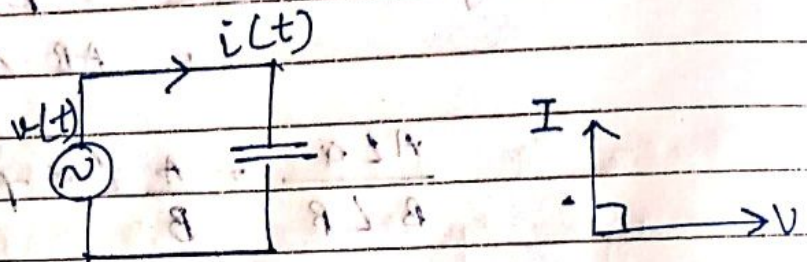
AC through pure capacitance -

$$q = CV_c$$

$$i(t) = \frac{dq}{dt} = C \frac{dV_m \sin \omega t}{dt}$$

$$= \frac{V_m \cos \omega t}{1/\omega C} = \frac{V_m}{X_c} \sin \left(\omega t + \frac{\pi}{2} \right)$$

$$X_c = \frac{1}{\omega C}$$

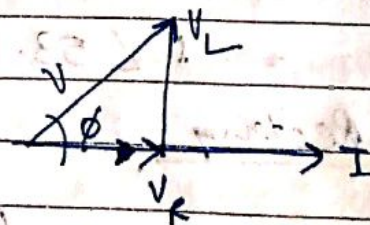
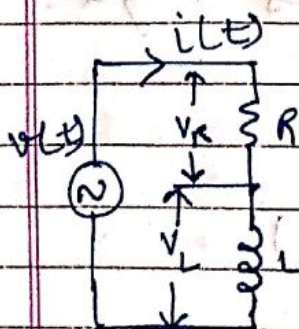


$$V(t) = V_m \sin \omega t$$

$$i(t) = I_m \sin \left(\omega t + \frac{\pi}{2} \right)$$

$$\begin{aligned} p(t) &= V(t) \cdot i(t) \\ &= V_m \sin \omega t I_m \sin \omega t \\ &= \frac{V_m I_m}{2} \sin 2\omega t \end{aligned}$$

$$P_{avg} = 0$$

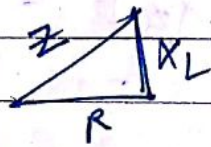


Phasor diagram

ϕ = Phase Angle

$$\phi = \tan^{-1} \left(\frac{V_m}{V_L} \right)$$

voltage & current are phasor (arrows will be there)
But impedance is not a phasor (no arrows)



impedance diagram.

$$v(t) = V_m \sin \omega t$$

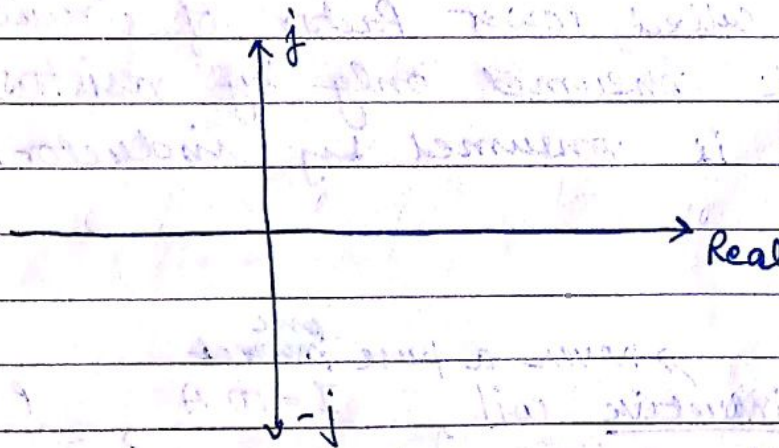
$$i(t) = \frac{V_m}{Z} \sin(\omega t - \phi)$$

$$\phi = \tan^{-1} \frac{V_L}{V_R} = \tan^{-1} \frac{X_L}{R}$$

$$\bar{V} = \bar{I} (R + jX_L)$$

$$j = \sqrt{-1}$$

it causes 90° phase shift.



Complex frequency (s) plane.

eg:- $j5 = 5 \angle 90^\circ$

$$\begin{aligned} p(t) &= V_m \sin \omega t I_m \sin(\omega t - \phi) \\ &= \frac{V_m I_m}{2} [\cos \phi - \cos(2\omega t - \phi)] \end{aligned}$$

$$= VI [\cos \phi - \cos(2\omega t - \phi)]$$

$$P_{avg} = VI \cos \phi$$

use I^2R . $\Rightarrow$ Avoid using $\frac{V}{R}$.

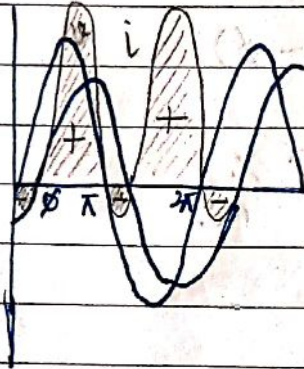
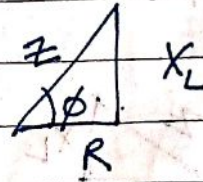
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$$P_{avg} = VI \cos \phi$$
$$= I Z I \frac{R}{Z}$$

$$= I^2 R$$

(Watts.)



$$P_{avg} = VI \cos \phi.$$

Q. $\cos \phi$ is called Power Factor of circuit And the power is consumed only by resistors & no power is consumed by inductor.

Q. In an inductive coil, $I = 10 \text{ A}$, $P = 1000 \text{ W}$
 $V = 250 \text{ V}$, $f_{req.} = 25 \text{ Hz}$

$\rightarrow$ never a pure ~~inductor~~ ^{one} ~~inductor~~
if not

Calculate i) Impedance

ii) Effective Resistance

iii) Reactance

iv) Power Factor

v) Phase angle.

Ans-

$$P = I^2 R$$

$\Rightarrow$

$$1000 = 10 \times 10 \times R$$

$$R = 10 \Omega$$

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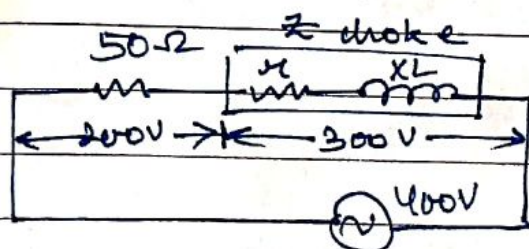
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Ques: When an AC voltage of 400 V is applied to resistor & a choke coil. The voltage across them are 200 V & 300 V respectively. ~~and current flowing to circuit is 4 A.~~ Find Power & P. Factor of circuit & choke coil. R

Resis. of 50 Ω is connected to choke coil.

Sol:



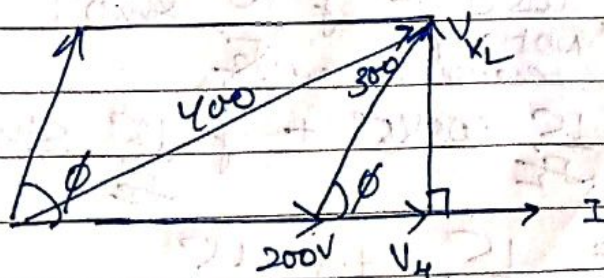
$$I = \frac{200}{50} = 4 \text{ A}$$

$$Z_{ckt} = \frac{400}{4} = 100$$

$$= \sqrt{(50 + R)^2 + (X_L)^2}$$

$$Z_{choke} = \frac{300}{4} = 75 = \sqrt{R^2 + X_L^2}$$

Through Phasor —

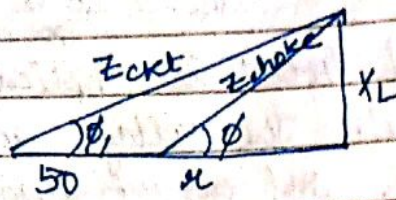


→ Phasor of R & X_L is 300

→ Phasor of 200 + 300 is 400.

$$200^2 + 300^2 + 2 \times 200 \times 300 \cos \phi = 400^2$$

$$\cos \phi = \frac{400^2 - 300^2 - 200^2}{2 \times 200 \times 300}$$



$$Z \cos \phi = R$$

$$Z \sin \phi = X_L$$

$$P_{ckt} = I^2 (50 + R)$$

$$P_{choke} = I^2 R$$

$$P.F._{ckt} = \frac{50 + R}{Z_{ckt}}$$

Q.2: 2 wires carrying I_1 & I_2 are connected to 3rd wire carrying I_3 . Find the expression for I_3 and also find rms value I_1 , I_2 , I_3 .

$$i_1(t) = 10 \sin \omega t.$$

$$i_2(t) = 15 \sin(\omega t + 45^\circ) + \cos 3\omega t.$$

rms value

$$\frac{10}{\sqrt{2}} \angle 0^\circ \rightarrow \frac{10}{\sqrt{2}} \cos 0^\circ + j \frac{10}{\sqrt{2}} \sin 0^\circ = \frac{10}{\sqrt{2}}$$

($\sqrt{2}$) NOT req. because we want peak.

$$\frac{15}{\sqrt{2}} \angle 45^\circ \rightarrow \frac{15}{\sqrt{2}} \cos 45^\circ + j \frac{15}{\sqrt{2}} \sin 45^\circ$$

$$= \frac{15}{\sqrt{2} \times \sqrt{2}} + j \frac{15}{\sqrt{2} \times \sqrt{2}}$$

Ans $I_3 = I_1(t) + I_2(t) = 23 (\sin \omega t + 27.2^\circ) + \cos 3\omega t.$

$$\left(10 + \frac{15}{\sqrt{2}} \right) + j \frac{15}{\sqrt{2}} = 20.6 + j 10.6$$

$$\tan^{-1} \left(\frac{10.6}{20.6} \right) = \approx 27.2^\circ$$

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• ACTIVE POWER (Real or True Power) -

In general in AC ckt there is phase diff. in b/w voltage & current phasors.

The express $P = VI \cos \phi$.

i.e. the component of current which is in phase with the voltage phasor gives rise to the active power.

This power is r/a active, real or True Power.

Also c/a wattful power.

Responsible for doing actual work like producing torque in motors, light etc.

measured using watt meter, (W).

At

In purely inductive & purely capacitive ckt there is phase diff. of 90° b/w current & voltage.

The net. power flows in both direction & reacts open itself w/out doing any actual work.

- The component of current 90° out of phase ($I \sin \phi$) with no voltage phasor is responsible for this reactive power also c/a Wattless power.

Unit is VAR (wattless).

• APPARENT POWER - $V_{rms} I_{rms}$

Unit is Volt ampere

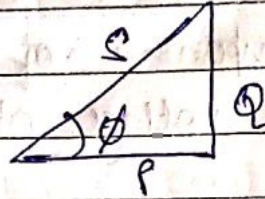
also c/a complex power.

complex power, $S = P \pm jQ$ $\rightarrow$ active power
 $\rightarrow$ Reactive Power.

active Power; $P = VI \cos \phi = I^2 R$.

Reactive Power; $Q = VI \sin \phi = I^2 X$.

Power Triangle -



For inductive loads, Q is +ve.

Capacitive loads, Q is -ve.

⊛ What is Power factor? And dis. of having low P.F?

$\rightarrow$ It is cosine of angle b/w volt. & current phasor.

$\rightarrow$ It is given by $\frac{R}{Z}$

$\rightarrow$ It is also given by $\frac{\text{active P.}}{\text{apparent P.}}$

$$P_f = \frac{\text{active Power}}{\text{Apparent Power.}}$$

Disadv. of having low P.f. $\rightarrow$

- $\rightarrow$ If Power & volt are const. low P.f means higher current which causes
 1. larger conductor size. for
 2. More efficiency becoz of more high Copper loss ($I^2 R$ losses)
 3. larger voltage drop. When the current is high

volt. drop is more. So, volt. regulation is poor.

$$V. \text{ Regulation} = \frac{V_{NL} - V_{FL}}{V_{FL}}$$

4. Large KVA rating of equipments.

The KVA rating of machines connected to power equipments such as transformers etc. will be more at low P.f.

Since it is $\propto \frac{1}{P.f.}$

$$KVA \propto \frac{1}{P.f.}$$

→ Capacitor, Capacitance banks & synchronous condensers (motors) can work in over excite state, for improving power factor.

Ques. - $v(t) = 250 \sin \omega t + 50 \sin(3\omega t + \pi/3) + 20 \sin(5\omega t + \pi/3)$

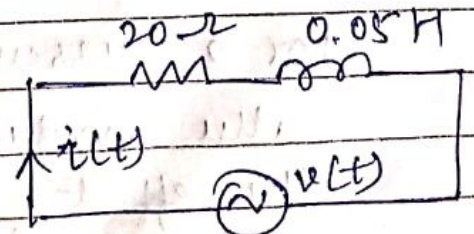
The $v(t)$ is applied to a circuit having $R = 20 \Omega$ & inductance $L = 0.05$ Henry.

Derive expression for I . Find I_{rms} & V_{rms} .
Total Power supplied & power factor.

Solⁿ -

$$Z(\omega) = \omega L$$

$$f_{req} = 50 \text{ Hz}$$



$$X(\omega) = \omega L = 2 \times 3.14 \times 50 \times 0.05 = 15.7$$

$$X(3\omega) = 3\omega L = 3 \times 2 \times 3.14 \times 50 \times 0.05 = 47.1$$

$$X(5\omega) = 5\omega L = 5 \times 2 \times 3.14 \times 50 \times 0.05 = 78.5$$

$$\begin{aligned}
 Z(\omega) &= 20 + jX(\omega) \\
 &= 20 + j15.7 \\
 &= 25.4 \angle 38.13^\circ
 \end{aligned}$$

$$\begin{aligned}
 Z(3\omega) &= 20 + jX(3\omega) \\
 &= 20 + j47.1 \\
 &= 51.18 \angle 67.8^\circ
 \end{aligned}$$

$$\begin{aligned}
 Z(5\omega) &= 20 + jX(5\omega) \\
 &= 20 + j78.5 \\
 &= 81 \angle 75.7^\circ
 \end{aligned}$$

$$i(t) = \frac{250}{25.43} \sin(\omega t - 38.13^\circ) + \frac{50}{51.18} \sin(3\omega t + 60^\circ)$$

$$v(t) = 250 \sin \omega t + 50 \sin(3\omega t - 60^\circ) + 20 \sin(5\omega t + 300^\circ)$$

$$\begin{aligned}
 i(t) &= \frac{250}{25.43} \sin(\omega t - 38.13^\circ) + \frac{50}{51.18} \sin(3\omega t + (60 - 67^\circ)) \\
 &\quad + \frac{20}{81.03} \sin(5\omega t + (300 - 75.7^\circ))
 \end{aligned}$$

$$\begin{aligned}
 &= 9.83 \sin(\omega t - 38.13^\circ) + 0.97 \sin(3\omega t - 7^\circ) \\
 &\quad + 0.247 \sin(5\omega t + 224.2^\circ)
 \end{aligned}$$

For a current having higher harmonic component the inductor offers higher impedance. The effect of inductance is to smoothen out of ripples in an AC waveform, so these are used in filters.

$$V_{rms} = \sqrt{\left(\frac{250}{\sqrt{2}}\right)^2 + \left(\frac{50}{\sqrt{2}}\right)^2 + \left(\frac{20}{\sqrt{2}}\right)^2}$$

$$I_{rms} = \sqrt{\left(\frac{9.83}{\sqrt{2}}\right)^2 + \left(\frac{0.97}{\sqrt{2}}\right)^2 + \left(\frac{0.247}{\sqrt{2}}\right)^2}$$

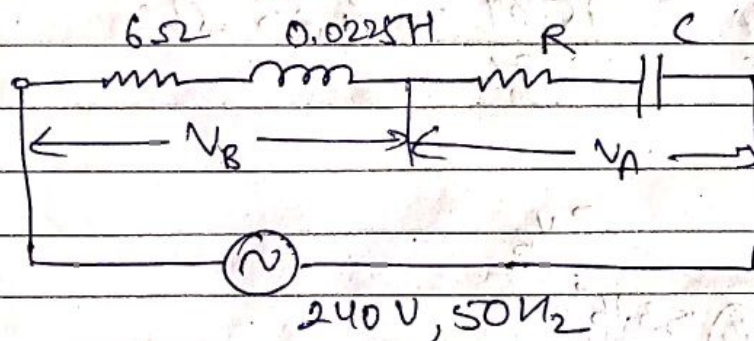
$$P = I^2 R$$

$$P.f = \frac{\text{active Power}}{\text{apparent Power}}$$

$$P.f = \frac{I^2 R}{VI}$$

30/01.

Ques:



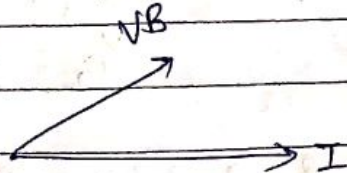
V_A & V_B are in phase quadrature (Phase angle of 90°)

$$V_B = 3V_A$$

Find R & C .

Soln

Ref. Phasor is always I .



$$V_B = 2 \times 3.14 \times 50$$

$$Z_B = \sqrt{6^2 + (3.14 \times 2 \times 50 \times 0.025)^2} \angle 53.13^\circ$$

$$Z_B = 10 \angle 53.13^\circ \Omega$$

$$V_A^2 + V_B^2 = (240)^2$$

$$V_A^2 + (3V_A)^2 = 240^2$$

$$10 V_A^2 = 240^2$$

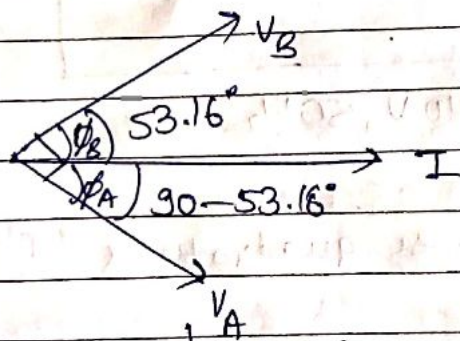
$$V_A = 75.89 \text{ V}$$

$$V_B = 3V_A = 3 \times 75.89$$

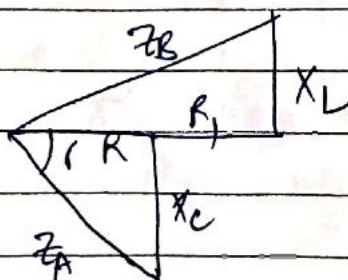
$$V_B = 227.68 \text{ V}$$

$$I = \frac{V_B}{Z_B} = \frac{227.68}{10} = 22.76 \text{ A}$$

$$Z_A = \frac{V_A}{I} = \frac{75.89}{22.76} = 3.33 \Omega$$



this can be other side bt. I is the leading the phasor.



$$R_1 = Z_A \cos \phi_A$$

$$X_L = Z_A \sin \phi_A$$

$$R = 3.33 \times \cos(90 - 53.16) = 2.6$$

$$X_L = 3.33 \times \sin(90 - 53.16)$$

Q. When an Resis + indu are in series & connect to 240 V, 50 Hz AC mains. ~~for~~ $I = 5A$ flows lagging 30° behind the supply voltage. While Vol. across inductor = 200 V.

Determine the resistance of resistor & reactance of inductor. Draw Phasor Diagram.

Q. A series ckt of 300Ω , non inductive resistor $2.59 \mu F$ capa. $2.36 H$ inductor of neg. resis. Supply - 250 V, 50 Hz. Calculate ckt current, phase angle & voltage drop across each elt.

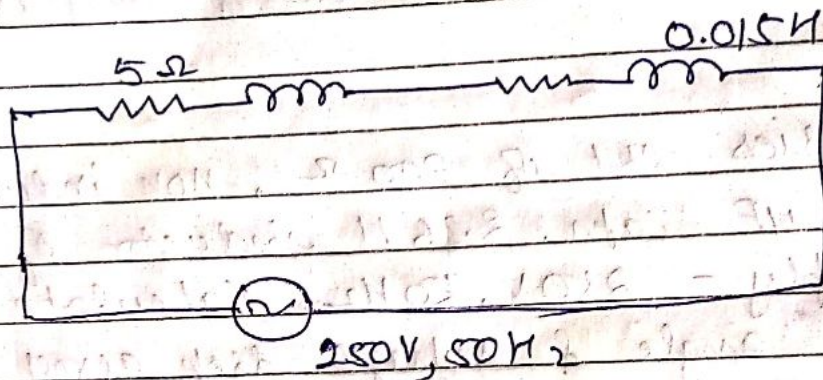
Q. An inductive coil having $R = 50 \Omega$ $I = 4A$ when connected to 100 V, 60 Hz supply. If coil is connected to 100 V, 50 Hz supply calculate current, Power & Power factor

Q. A $v_L(t) = 100 \sin(314t)$ is applied to a series ckt having 100Ω resistance $L = 0.0318 H$. capacitance $63.6 \mu F$. Find the expression for ~~is~~ $i(t)$, phase angle, Power factor

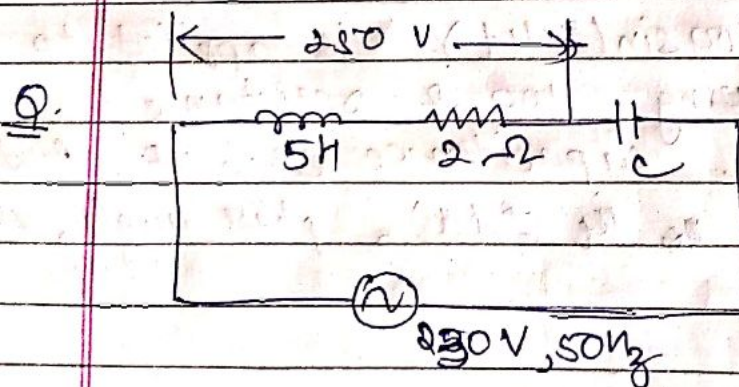
Q. When res. & inductor in series are connected to 240 V, 50 Hz AC mains. A current of 3 A flows lagging 37° behind supply voltage. While voltage across ind. is 171 V. Det. the resistance of the resistor, reactance of inductance. Draw phasor Diagram.

(2015 mid-sem)

Q 2 coils A & B are in series across 250V, 50Hz supply. The resist. of A is 5Ω & ind. of B is $0.015H$. Input from supply is $4kW$ & $3KVAR$. Find inductance of A & resistance of B. Also find voltage across each coil.



$$\cos \phi = \frac{KW}{KVAR} = \frac{4}{3}$$



What should be the value of C so that voltage across each is 250V

$$I = \frac{250}{Z_{coil}}$$

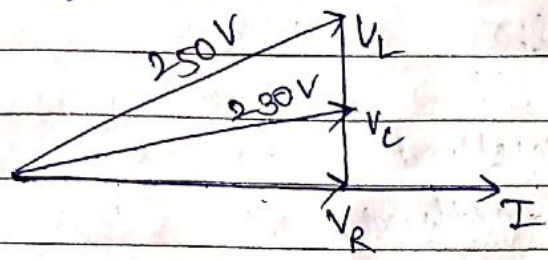
$$Z_{ckt} = \frac{250}{I} = \sqrt{R^2 + (X_L - X_C)^2}$$

$$X_L = \omega L = 2\pi f \times L = 2 \times 3.14 \times 50 \times 5 = 1570$$

- Iron losses are fixed losses
- Copper losses are variable losses

Q

same



$$Z_{\text{coil}} = \sqrt{2^2 + (2 \times 3.14 \times 50 \times 5)^2} \angle 89.9^\circ$$

$$= 1570 \angle 89.9^\circ \Omega$$

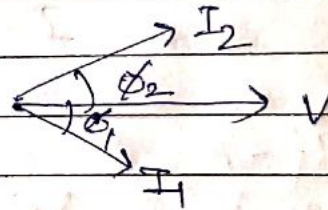
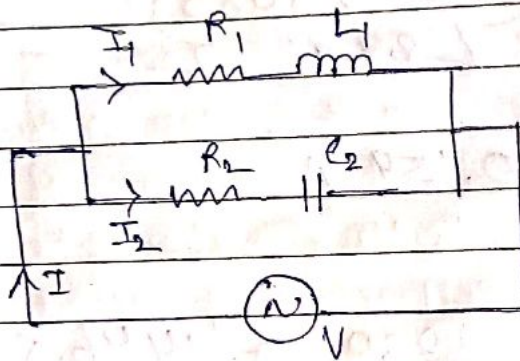
$$I = \frac{250}{1570} = 0.159 \text{ A}$$

$$Z_{\text{ckt}} = \frac{230}{I} = \frac{230}{0.159} = 1446.5 \Omega$$

02.

Parallel AC Ckt:-

- ① Phasor diagram method
- ② Complex Algebra method.



in Capacitor branch - current will be leading the voltage.

in inductor branch - current will be lagging.

$$\bar{V} = V \angle 0^\circ$$

$$\bar{Z}_1 = R_1 + j\omega L_1 = |Z_1| \angle \phi_1$$

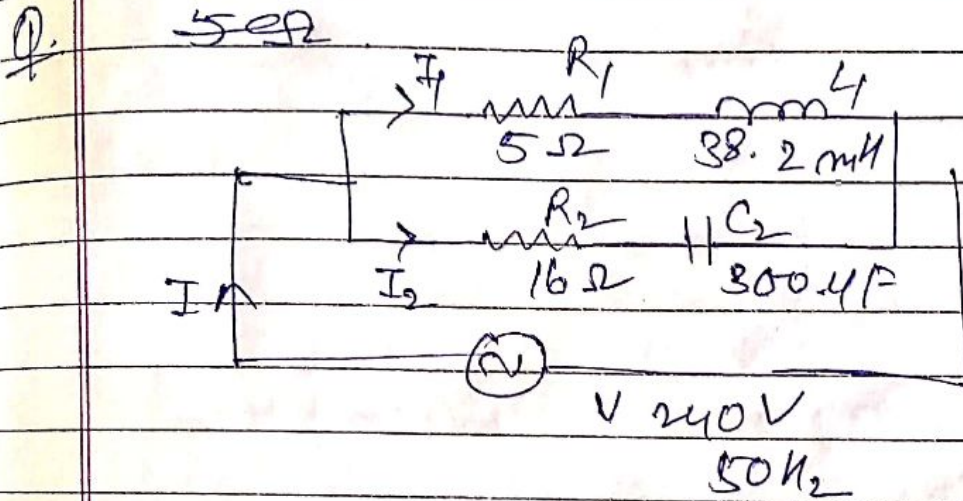
$$\bar{Z}_2 = R_2 - j(\frac{1}{\omega C_2}) = |Z_2| \angle -\phi_2$$

$$\bar{I}_1 = \frac{V \angle 0^\circ}{Z_1 \angle \phi_1} = |I_1| \angle -\phi_1$$

$$\bar{I}_2 = \frac{V \angle 0^\circ}{Z_2 \angle -\phi_2} = |I_2| \angle \phi_2$$

$$\vec{I} = \vec{I}_1 + \vec{I}_2 = 17.6 \angle 0^\circ$$

$$Z_{eq} = \frac{V \angle 0^\circ}{I \angle 0^\circ}$$



$$\vec{Z}_1 = 5 + j 2 \times 3.14 \times 50 \times 38.2 \times 10^{-3}$$

$$= 5 + j 11.99$$

$$= 12.99 \angle 67.37^\circ$$

$$\vec{Z}_2 = R_2 - j \frac{1}{\omega C} = 16 - j \frac{1}{2 \times 3.14 \times 50 \times 300 \times 10^{-6}}$$

$$= 16 - j \frac{1}{94,200} \times 10^6 = 16 - j 1.061 \times 10^{-5} \times 10^6$$

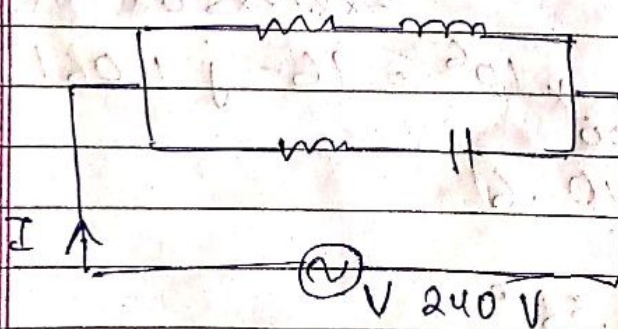
$$= 16 - j 10.61$$

$$= 19.19 \angle -33.55^\circ$$

$$\vec{I}_1 = \frac{240 \angle 0^\circ}{12.99 \angle 67.37^\circ} = 18.47 \angle -67.37^\circ$$

$$\vec{I}_2 = \frac{240 \angle 0^\circ}{19.19 \angle -33.55^\circ} = 12.5 \angle 33.55^\circ$$

Ques: ~~a~~ ~~amp~~ $Z_1 = 10 + j15$, $Z_2 = 6 - j8$
 are connected in parallel. Total current = 10A
 What is the Power taken by each branch.



$$\bar{I} = I \angle 0^\circ$$

$$\begin{aligned} \bar{I} &= \frac{\bar{I} \times \bar{Z}_2}{\bar{Z}_1 + \bar{Z}_2} = \frac{10 \angle 0^\circ \times 10 \angle -53.12^\circ}{16 + j7} \\ &= \frac{100 \angle -53.12^\circ}{17.46 \angle 23.6^\circ} \end{aligned}$$

$$I_2 = \frac{\bar{I} \times Z_1}{Z_1 + Z_2} = \frac{10 \angle 0^\circ \times 18.02 \angle 56.3^\circ}{16 + j7} = \frac{180.2 \angle 56.3^\circ}{17.46 \angle 23.6^\circ}$$

$$P_1 = |I_1|^2 R_1$$

$$P_2 = |I_2|^2 R_2$$

Q. 2 impedencing having same num. value.
one having P.f. = 0.6, other having P.f. = 0.8
are connected in parallel. Find P.f. of equivl.
ckt.

Sol. Both inductive.

$$Z_1 = Z \angle \cos^{-1} 0.6 = Z \angle 53.13^\circ$$

$$Z_2 = Z \angle \cos^{-1} 0.8 = Z \angle 36.86^\circ$$

$$Z_{eq} = \frac{\bar{Z}_1 \bar{Z}_2}{Z_1 + Z_2} = \frac{Z \angle 53.13^\circ Z \angle 36.86^\circ}{Z[0.6 + j0.8 + 0.8 + j0.6]}$$

$$Z_{eq} = \frac{Z^2 \angle 90^\circ}{Z[1.4 + j1.4]} = \frac{Z \angle 90^\circ}{1.67 \angle 45^\circ} = \frac{Z \angle 45^\circ}{1.67}$$

$$P.f. = \frac{R}{Z}$$

Admittance Method

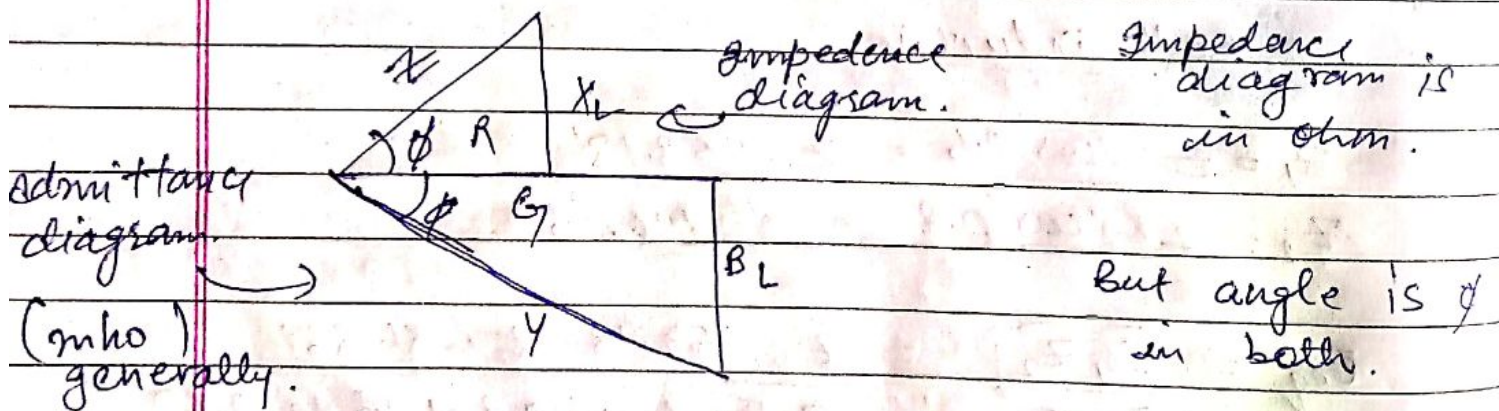
$$\bar{Y} = \frac{1}{\bar{Z}} = G \mp jB.$$

$\downarrow$ conductance $\rightarrow$ susceptance

$\Rightarrow$ R-L ckt.

$$\bar{Y} = \frac{1}{(R + jX_L)} \cdot \frac{(R - jX_L)}{(R - jX_L)} = \frac{R}{R^2 + X_L^2} - \frac{jX_L}{R^2 + X_L^2}$$

$$= \frac{R}{Z^2} - j \frac{X_L}{Z^2}$$

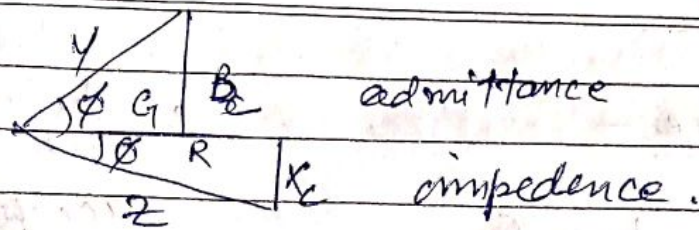


(R-L ckt.)

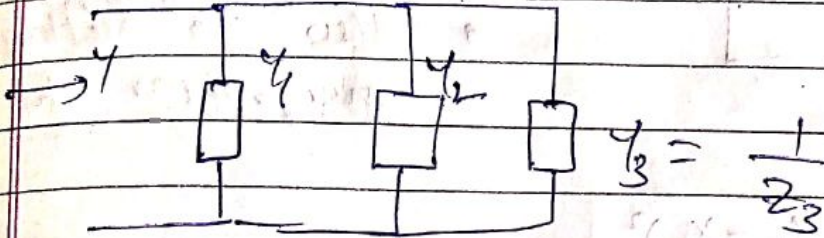
$\Rightarrow$ R-C ckt.

$$\bar{Y} = \frac{1}{(R - jX_C)} \cdot \frac{(R + jX_C)}{(R + jX_C)} = \frac{R}{R^2 + X_C^2} + j \frac{X_C}{R^2 + X_C^2}$$

$$= \frac{R}{Z^2} + j \frac{X_C}{Z^2}$$



Y - admittance.



$$Y = Y_1 + Y_2 + Y_3$$

Ques:

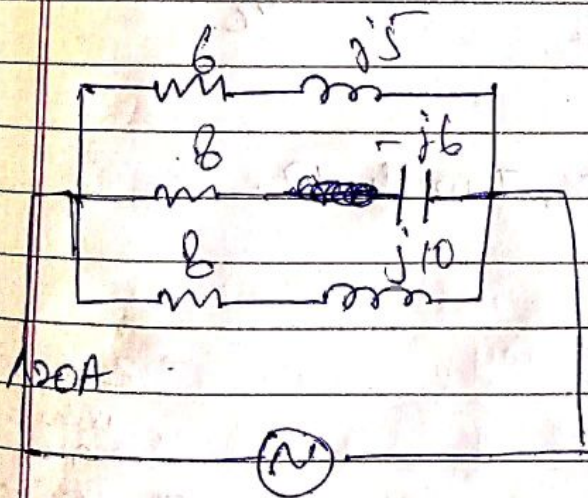
Total current = 20A.

$$Z_1 = 6 + j5$$

$$Z_2 = 8 - j6$$

$$Z_3 = 8 + j10$$

Find current in each branch.



$$V = \frac{I}{Y}$$

$$V = IZ$$

$$I = \frac{V}{Z} = VY$$

$$I_1 = VY_1, I_2 = VY_2$$

$$Y = I_1 + I_2 + I_3$$

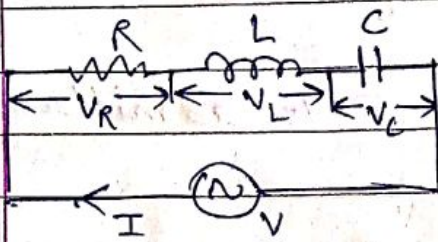
$$Y_1 = \frac{6}{6^2 + 5^2} - \frac{j5}{6^2 + 5^2}$$

$$Y_2 = \frac{8}{8^2 + 6^2} + \frac{j6}{8^2 + 6^2}$$

$$Y_3 = \frac{8}{8^2 + 10^2} - \frac{j10}{8^2 + 10^2}$$

06/02.

(*) Resonance:-



- Series Resonance ckt is also c/a acceptor ckt becoz ckt current is max.
- series resonance, also c/a voltage magnifier ckt.

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$Z_0 = R$$

$$|V_L| = |V_C|$$

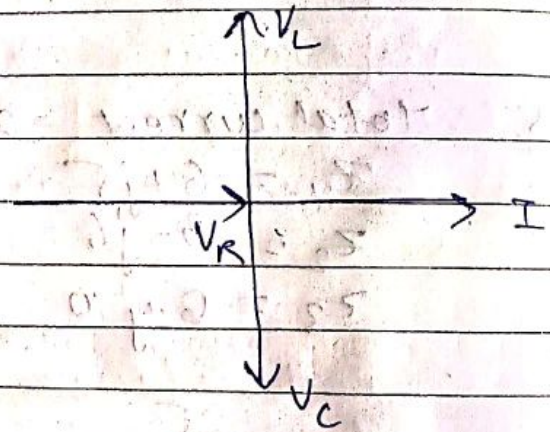
$$IX_L = IX_C$$

$$\omega L = \frac{1}{\omega C}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

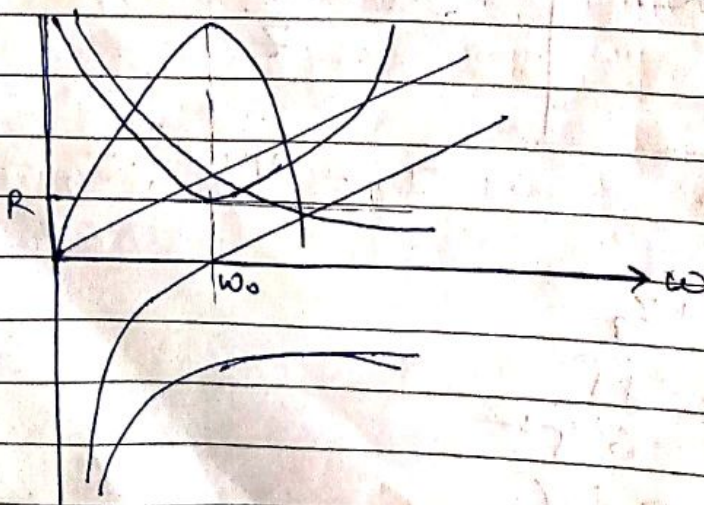
$\omega = \omega_0$

resonance angular freq.



→ Variation of X_L, X_C, Z, I with ω .

ques. comes. give explanation also.



- Z min while resonance
- ckt. current will be max.

Series resonance may be produced by varying frequency or by varying L & C or both.

In practice, it is convenient to vary resonance by changing capacitance.

Quality Factor :-

$$Q = \frac{2\pi \times \text{energy stored per cycle}}{\text{energy dissipated per cycle}} \quad \rightarrow \text{while resonance.}$$

$$W_{so} = \frac{1}{2} L i^2(t) + \frac{1}{2} C v^2(t)$$

$$\text{let } i(t) = I_m \sin \omega t$$

$$v(t) = \frac{1}{C} \int i(t) dt$$

$$v(t) = \frac{1}{C} \int I_m \sin \omega t dt = \frac{I_m}{\omega C} \cos \omega t$$

$$W_{so} = \frac{1}{2} L I_m^2 \sin^2 \omega t + \frac{1}{2} \frac{I_m^2 \cos^2 \omega t}{\omega_0^2 C}$$

$$= \frac{1}{2} L I_m^2 [\sin^2 \omega t + \cos^2 \omega t]$$

$$\boxed{\omega_0^2 = \frac{1}{LC}}$$

I_m
↓
peak
value
at reso.

$$W_{so} = \frac{1}{2} L I_m^2 = L I_0^2 \quad \left[I_0 = \frac{I_m}{\sqrt{2}} \right]$$

also

$$W_{do} = I_0^2 R T = I_0^2 \frac{R}{f_0} = I_0^2 \frac{R \times 2\pi}{\omega_0}$$

$$Q = \frac{2\pi \times L I_0^2 \times \omega_0}{2\pi I_0^2 R} = \frac{\omega_0 L}{R} \quad \text{or} \quad \frac{1}{\omega_0 R C}$$

$$\text{So, } \# \quad \boxed{Q = \frac{\omega_0 L}{R} \quad \text{or} \quad \frac{1}{\omega_0 R C}}$$

The quality fac. is also a measure of voltage magnification in RLC series ckt.

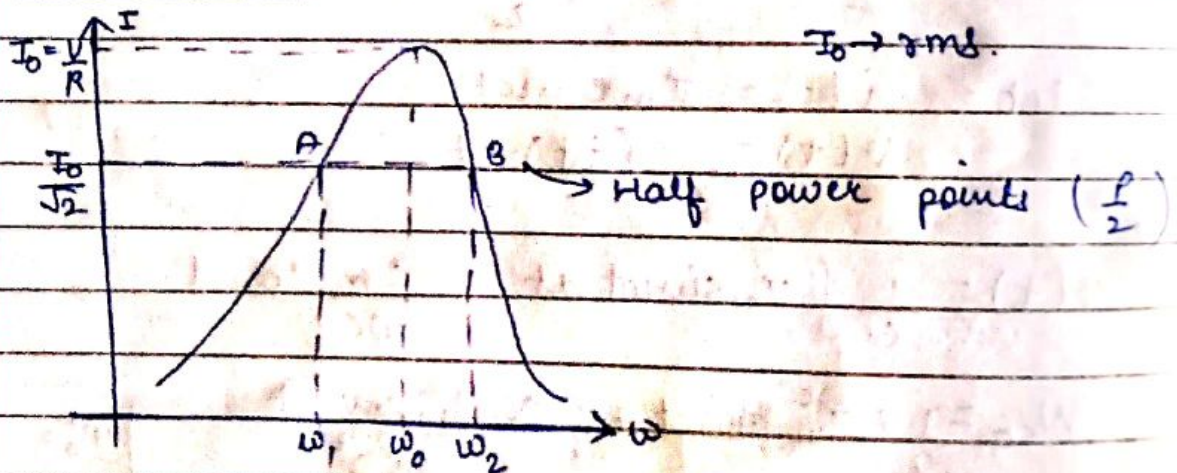
$$V = V_R$$

★ Voltage magnification,

$$Q = \frac{V_{L_0}}{V} = \frac{V_{L_0}}{V_R} = \frac{\omega_0 L}{R} = \frac{\omega_0 L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

also, $Q = \frac{1}{R} \sqrt{\frac{L}{C}}$

★ Bandwidth:-



In series reso. ckt, circuit current is max at resonance frequency.

Bandwidth is the range of frequencies which lies ^{4/10 2 pts on} either side of resonating freq. where current is $\frac{I_0}{\sqrt{2}}$

$$I_0 = \frac{V}{R}$$

$$I(\omega_2) = I(\omega_1) = \frac{I_0}{\sqrt{2}}$$

$$I = \frac{V}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

$$I(\omega_1) = I(\omega_2) = \frac{I_0}{\sqrt{2}} = \frac{V}{\sqrt{2}R}$$

$$\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} = \sqrt{2}R$$

sq. both sides

$$R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2 = 2R^2$$

$$\left(\omega L - \frac{1}{\omega C}\right)^2 = R^2$$

$$\left(\omega L - \frac{1}{\omega C}\right) = \pm R$$

$$\rightarrow \omega^2 LC - 1 \pm R\omega C = 0$$

This relation shows that the mag. of reactance of Z is equal to resistive component of Z .

At the upper & lower cut of freq. the reactance will be equal to resistance & the phase angle will be 45° hence power factor is $\frac{1}{\sqrt{2}}$.

$$\omega_2 L - \frac{1}{\omega_2 C} - R = 0$$

$$\omega_1 L - \frac{1}{\omega_1 C} + R = 0$$

$$\Rightarrow \omega_1 = \frac{-R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\Rightarrow \omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

Q is selective measure of bandwidth.

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$$\omega_2 - \omega_1 = \frac{R}{2L} - \left(-\frac{R}{2L} \right)$$

$$\boxed{\omega_2 - \omega_1 = \frac{R}{L}} \quad \text{Bandwidth.}$$

$$\text{Let } \frac{R}{2L} = \alpha$$

$$\omega_1 = -\alpha + \sqrt{\alpha^2 + \omega_0^2}$$

$$\omega_2 = \alpha + \sqrt{\alpha^2 + \omega_0^2}$$

$$\omega_1 \omega_2 = \alpha^2 + \omega_0^2 - \alpha^2 = \omega_0^2$$

$$\boxed{\omega_0 = \sqrt{\omega_1 \omega_2}} \quad \text{Resonant freq.}$$

These eqⁿ show that a bandwidth of series RLC ckt depends on ratio of R & L and is independent of C. But value of ω_0 is dependent on C.

$$\omega_2 - \omega_1 = \frac{R}{L}$$

$$Q = \frac{\omega_0 L}{R} = \frac{\omega_0}{\omega_2 - \omega_1} = \frac{f_0}{f_2 - f_1} \quad \text{selectivity.}$$

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 RC} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

SELECTIVITY

⇒ For a given resonant frequency higher 'Q' means smaller bandwidth and sharper the curve, —
Such a ckt is said to have a greater freq. selectivity.

The pts. A & B are also c/a half power pts.
 i.e. power dissipated at A & B is half that of
 power dissipated at resonant frequency.

$$10 \log_{10} \frac{\text{Power at A (or B)}}{\text{Power at resonance}}$$

$$= 10 \log_{10} \frac{\left(\frac{I_0}{\sqrt{2}}\right)^2 R}{I_0^2 R} = -3 \text{ dB.}$$

$$\omega_1 = \omega_0 - \frac{\omega_2 - \omega_1}{2} \quad ; \quad \omega_2 = \omega_0 + \frac{\omega_2 - \omega_1}{2}$$

Ques: A coil of $R = 40 \Omega$, $L = 0.75 \text{ Henry}$,
 forms part of series ckt for reso. freq. $\approx 55 \text{ Hz}$
 when $V = 250 \text{ V}$, 50 Hz Find line current,
 P.f., and vol. across coil.

Ques: The $Q = 5$, reso. freq. = 1 kHz :
 assuming power dissipated = 250 Watts , when ω is
 Find circuit parameters & bandwidth.

Sol:

$$Q = \frac{f_0}{f_2 - f_1}$$

$$\omega_2 - \omega_1 = \frac{R}{L}$$

$$I^2 R = P$$

$$f_0 = 1 \text{ kHz}$$

$$f_2 - f_1 = 1000 = 200$$

$$200 \times \omega = 200 \times 5 = 1256 \text{ Hz}$$

$$250 = I^2 R$$

$$R = 250$$

$$R = 250 \Omega$$

$$L = 0.190 \text{ H}$$

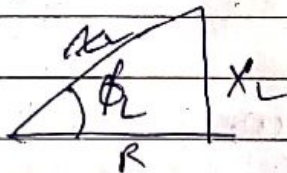
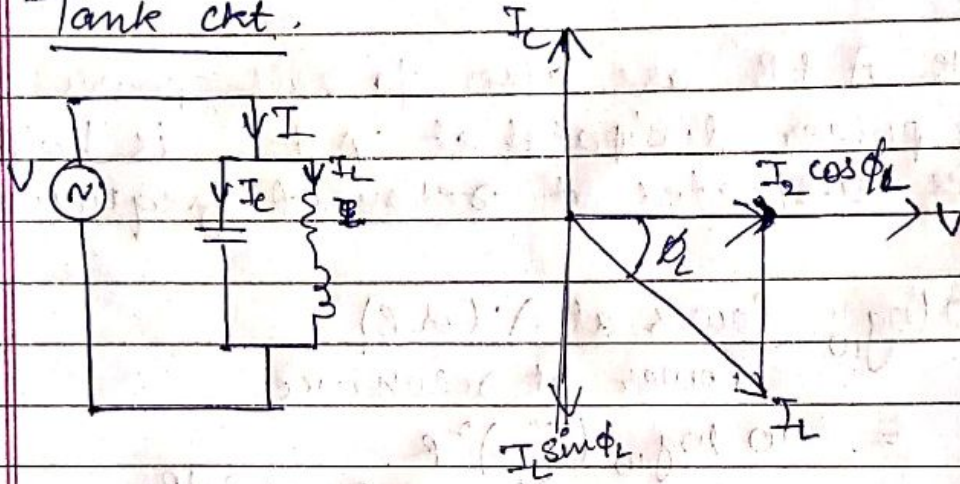
$$C = 0.12 \mu\text{F}$$

$$\text{bandwidth} = 1256 \text{ Hz}$$

07/02

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Parallel Resonance Circuit:-Tank ckt.

Dynamic impedance.

$$* \begin{cases} I_0 = I_L \cos \phi_L \\ I_C = I_L \sin \phi_L \end{cases}$$

→ value of ckt impedance is max. at reson.

$$\frac{V}{X_C} = \frac{V}{Z_L} = \frac{V}{Z_L}$$

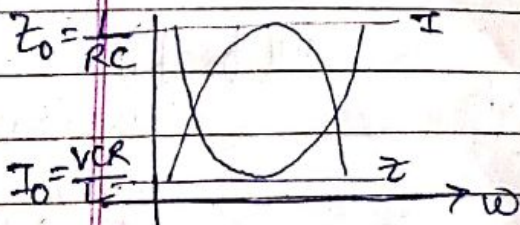
→ value of ckt current is min. at resonance.

$$Z_L^2 = X_L X_C = \frac{\omega L}{\omega C}$$

$$I_0 = \frac{V}{Z_0} = \frac{V}{Z_L} \cdot \frac{R}{Z_L} = \frac{VR}{Z_L^2}$$

$$[I_0 = \frac{VCR}{L}]$$

$$[Z_0 = V/I_0 = L/RC] \Rightarrow \text{Dynamic Impedance} (*)$$



$$Z_L^2 = \frac{L}{C} = R^2 + \omega_0^2 L^2$$

$$\omega_0 = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}} \quad R \ll L \Rightarrow \boxed{\omega_0 \propto \frac{1}{\sqrt{LC}}}$$

Since the ckt current is small $= (VCR)/L$, it is c/a rejector ckt.

If R is negligible then ckt current is also negligible. branch current I_L & I_C are very high becoz of transfer of energy from inductor to capacitor & vice-versa. It is basis for oscillator.

The energy source is only needed to supply for the losses in the resistance.

This ckt is also called current magnifier ckt.

⊛ Current Magnification:

$$= \frac{I_C}{I_0} = \frac{I_L \sin \phi_L}{I_L \cos \phi_L}$$

$$= \tan \phi_L = \frac{\omega_0 L}{R}$$

X

X

X